COGS 181A - Homework 1

due Tuesday, January 13 at 11:59PM

Problem 1. (10 points) Conceptual Questions

Assume we have a binary classification model:

$$f(x) = \begin{cases} +1, & \text{if } \mathbf{w} \cdot \mathbf{x} + b \geq 0 \\ -1, & \text{if } \mathbf{w} \cdot \mathbf{x} + b < 0 \end{cases}$$

where the feature vector $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$, bias $b \in \mathbb{R}$, weight vector $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$. The decision boundary of the classification model is: $\mathbf{w} \cdot \mathbf{x} + b = 0$

1.1 If the predictions of the classifier f and its decision boundary $\mathbf{w} \cdot \mathbf{x} + b = 0$ are shown in Figure 1, which one below can be a possible solution of weight vector $\mathbf{w}$ and bias b?

A. $\mathbf{w} = (+1, 0), b = -1$.

B. $\mathbf{w} = (-1, 0), b = +1$.

C. $\mathbf{w} = (+1, 0), b = +1$.

D. $\mathbf{w} = (0, -1), b = -1$.

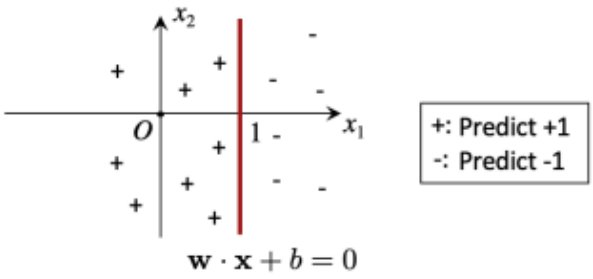


Figure 1: Figure 1

1.2 What is the decision boundary of the classification model?